

Lifesigns Protocol (v1.1)

1、Bluetooth Service Information (UUIDs):

Comm Service:	49535343-FE7D-4AE5-8FA9-9FAFD205E455
Send Characteristic:	49535343-1E4D-4BD9-BA61-23C647249616
Receive Characteristic:	49535343-8841-43F4-A8D4-ECBE34729BB3
Rename Characteristic:	00005343-0000-1000-8000-00805F9B34FB
MAC Address Characteristic:	00005344-0000-1000-8000-00805F9B34FB

PS: Host application should use the notification of the 'Send Characteristic' for package fetching.

2、Device Packet (Device to Host):

Packet length:	207 bytes
Packet rate:	about 4.17Hz (240ms interval)
Packet Format:	

Byte0: 0xff	Head mark1
Byte1: 0xaa	Head mark2
Byte2: PktIndex	Packet index
Byte3: Status	Measurement status info
Byte4: SpO2Sat	Averaged blood oxygen saturation
Byte5: SpO2SatReal	Real time blood oxygen saturation
Byte6: PulseRate	Averaged pulse rate (i.e. heart rate)
Byte7: PulseRateReal	Real time pulse rate (i.e. heart rate)
Byte8: PerfuIndex	Averaged blood perfusion index
Byte9: PerfuIndexReal	Real time blood perfusion index
Byte10: ADCSample[0]	Lowest byte of signed int32 ADCSample
Byte11: ADCSample[0]	Lower byte of signed int32 ADCSample
Byte12: ADCSample[0]	Higher byte of signed int32 ADCSample
Byte13: ADCSample[0]	Highest byte of signed int32 ADCSample
Byte14: ADCSample[1]	Lowest byte of signed int32 ADCSample
Byte15: ADCSample[1]	Lower byte of signed int32 ADCSample
Byte16: ADCSample[1]	Higher byte of signed int32 ADCSample
Byte17: ADCSample[1]	Highest byte of signed int32 ADCSample
...	
...	
...	
Byte198: ADCSample[47]	Lowest byte of signed int32 ADCSample
Byte199: ADCSample[47]	Lower byte of signed int32 ADCSample
Byte200: ADCSample[47]	Higher byte of signed int32 ADCSample
Byte201: ADCSample[47]	Highest byte of signed int32 ADCSample
Byte202: Axis-X	X-channel acceleration value

Byte203: Axis-Y	Y-channel acceleration value
Byte204: Axis-Z	Z-channel acceleration value
Byte205: Battery	Battery power in percent
Byte206: Checksum	Checksum of the whole packet
PktIndex:	Range (0~255), the index of the current packet
Status:	0x01 SpO2 Sensor Off
	0x02 No finger
	0x04 No pulse signal
	0x08 Pulse Beat Tips
SpO2Sat:	Range (0~100, invalid value=127), unit: %
SpO2SatReal:	Range (0~100, invalid value=127), unit: %
PulseRate:	Range (25~250, invalid value=255), unit: bpm
PulseRateReal:	Range (25~250, invalid value=255), unit: bpm
PerfuIndex:	Range (1~200, invalid value=0), unit: ‰
PerfuIndexReal:	Range (1~200, invalid value=0), unit: ‰
ADCSample[0]:	Range ($-2^{31} \sim 2^{31}$), signed int32, The first infrared light adc sample
...	
ADCSample[47]:	Range ($-2^{31} \sim 2^{31}$), signed int32, The 48th infrared light adc sample
Axis-X	X-channel acceleration value of Triaxial acceleration sensor Range(-128~127), signed int8
Axis-Y	Y-channel acceleration value of Triaxial acceleration sensor Range(-128~127) , signed int8
Axis-Z	Z-channel acceleration value of Triaxial acceleration sensor Range(-128~127) , signed int8
Battery:	Range (0~100), unit: %
Checksum	Range (0~255) Formula: Checksum = (Byte0+Byte1+...+Byte206) % 256, % is mod operator

3、Host Command (Host to Device):

Command length: 1 byte

Command type:

0xf0 ~0xf3 ----- Reserved
0xf4 ----- ADCSample is Original sampling waveform
0xf5 ----- ADCSample is Filtered sampling waveform
0xf6 ----- Reserved
0xf7 ----- Automatic timed wake-up intervals: 1minute
0xf8 ----- Automatic timed wake-up intervals: 3minutes

0xf9 ----- Automatic timed wake-up intervals: 5minutes

0xff ----- Get Software Version

0xfe ----- Get Hardware Version

Software Version Package Example (20 bytes):

0xff 0xaa 0x53 0x56 0x31 0x2e 0x30 0x34 0x2e 0x30 0x30 0x2e 0x33 0x36
0x00 0x00 0x00 0x00 0x00 0x3a
0xff 0xaa ----- Packet head mark
0x53 ----- ASCII 'S', mark of software version
0x56 0x31 0x2e 0x30 0x34 0x2e 0x30 0x30 0x2e 0x33 0x36 ----- ASCII string
'V1.04.00.36', current software version content
0x00 0x00 0x00 0x00 0x00 ----- Padding bytes of packet
0x3a ----- Checksum byte

Hardware Version Package Example:

0xff 0xaa 0x48 0x56 0x32 0x2e 0x30 0x00 0x00 0x00 0x00 0x00 0x00 0x00
0x00 0x00 0x00 0x00 0x00 0xd7
0xff 0xaa ----- Packet head mark
0x48 ----- ASCII 'H', mark of hardware version
0x56 0x32 0x2e 0x30 ----- ASCII string 'V2.0', current hardware version content
0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 -----
Padding bytes of packet
0xd7 ----- Checksum byte

Demo source code:

Android:

BLE demo: <https://github.com/zh2x/SpO2-BLE-for-Android>

iOS:

Swift demo: <https://github.com/zhuchengji-berry/BluetoothDemo>

Flutter:

Dart demo: https://github.com/mz-myfc/bci_demo